

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: EUDM Time Sync Replenish Timeout objection is not relevant for UFP

Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Enhanced Uni-Directional Time Sync Replenish Timeout objection is relevant for DFP. The current description applies it for both UFP and DFP. This ECR change it to be DFP only objection.

Benefits as a result of the proposed:
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Spec clarification

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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NA

An analysis of the hardware implications:
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NA

An analysis of the software implications:
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NA

An analysis of the compliance testing implications:
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NA

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Actual Change

(a). 4.2.1.6.3 Objections

To Text:

- When using Enhanced Uni-Directional Time Sync handshake:
 - While using Inversed Bi-Directional mode.
 - The Downstream Facing Port sent a Delay Response and the Follow-Up Packet has not yet been sent.
 - The Downstream Facing Port missed *Replenish Timeout* number of Time Sync handshakes and has not yet completed a Time Sync handshake.
 - The Downstream Facing Port missed more than *Replenish Threshold* Time Sync handshakes, the Lane Adapter is not in CL2, CL1, or CL0s (Tx) state, and the TMU has not yet completed *ReplenishN* consecutive handshakes.

(b). 8.2.2.2 TMU Adapter Configuration Capability

To Text:

Table 8-11 TMU Adapter Configuration Capability Fields

DW	Register Name	Bit(s)	Field Name and Description	Type	Default Value
8	TMU_ADP_CS_8	14:0	Replenish Timeout This field contains the max <i>AdapterTimeSyncInterval</i> rate for which the <u>DFPTMU</u> will not object to CL1. If an Adapter is in CL1 state for more than this time (<i>AdapterTimeSyncInterval</i> * <i>Replenish Timeout</i>), the <u>DFPTMU</u> shall assert an objection to CL1 state.	R/W	0
		15	Enable Enhanced Uni-Directional Mode (EUDM) When this bit is set to 1b, the USB4 Port shall use Enhanced Uni-Directional Time Sync Handshakes.	R/W	0
		25:16	Replenish Threshold This field contains the number of missed Time Sync Handshakes that requires an objection to CL1 state. If more than <i>Replenish Threshold</i> handshakes are missed, the TMU shall assert an objection to CLx until it finishes at least <i>ReplenishN</i> handshakes. Shall be set to a lower value than in the <i>Replenish Timeout</i> field.	R/W	0
		31:26	Reserved	RO	0